

Predicting District Heating Networks Fault Location with Graph Neural Networks

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Abstract

District heating networks (DHNs) are critical infrastructure prone to physical failures such as leaks, which cause significant energy and financial losses. Traditional physics-based monitoring methods are computationally expensive and require the continual recalibration of complex mathematical models, while standard data-driven approaches often fail due to the scarcity of real-world sensor data. This study addresses these challenges by proposing a topology-aware graph neural network (GNN) architecture for fault localization. The methodology follows a two-stage process: first, a graph attention-based architecture is designed and optimized using a synthetic dataset to effectively capture multi-step neighborhood dependencies. Second, the model is adapted and evaluated on a physically simulated dataset of a real urban DHN, comprising 187 nodes and 42,570 operational states. The problem is formulated as a multi-class classification task across supply and return subnets. The results demonstrate high predictive performance, achieving an accuracy of 96% on the supply subnet and 91% on the return subnet. Analysis of prediction errors reveals a strong bias towards local topological mistakes, indicating the model's ability to capture the physical propagation of disturbances. These findings highlight the efficacy of GNNs in handling sparse data and exploiting network topology for robust DHN monitoring.

Keywords: district heating; graph neural networks; fault detection; leak localization; thermohydraulic simulation; network topology

1. Introduction

District heating is a vital component of urban infrastructure, supplying heat to dense urban populations worldwide. Like all infrastructure, it is susceptible to physical aging, corrosion, thermal stresses, and external impacts, leading to loss of insulation, pipe deformation, and reduced reliability of heat supply [1–3]. These changes disrupt the network's operation and ultimately result in failures. Such incidents cause not only substantial heat loss but also significant financial damage. Consequently, the detection and localization of leaks in district heating pipelines is a key operational priority. This challenge, however, is not unique to district heating systems.

When examining fault detection, one observes a fundamentally similar class of diagnostic problems appearing across sectors such as gas [4], water [5], and power [6] distribution. However, oil/gas and water distribution networks have a significantly longer

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and broader industrial history. Consequently, many established leak detection methodologies and their practical applications have evolved primarily in these sectors, leaving district heating networks (DHNs) with less developed or adapted techniques [7]. There are currently three main approaches to underground fault detection: physics-based methods [8], data-driven methods [9], and the analysis of infrared images captured by unmanned aerial vehicles (UAVs) [10].

Physics-based methods entail building a comprehensive, complex physical model, running numerical simulations, and comparing the model's predictions with real-world data (such as SCADA, or Supervisory Control and Data Acquisition, measurements). Differences between simulation results and reality offer limited options; effectively, the entire physical model must be reconstructed or recalibrated. This is a sufficiently complex and demanding process. Moreover, the model must either be intricate enough to account for external factors, corrosion, and measurement noise, or its continual recalibration will become a regular activity [11].

To avoid constantly updating physical models to evolving operational data, a direct utilization of such data (for instance, as multivariate time series) offers an alternative way for fault prediction. This approach gives rise to data-driven machine learning methods [12], which apply mathematical statistics to establish correlations between monitoring data and defect occurrences. Moreover, a wide range of modeling techniques is employed in this field, including gradient boosting [13], convolutional neural networks [14], physics-informed neural networks [15], each introducing distinct assumptions and inductive biases. Nevertheless, the effectiveness of these methods is highly dependent on data quality, particularly regarding sensor placement density, spatial distribution, and measurement frequency (with some approaches requiring the deployment of extra sensors or the collection of external measurements [16]).

However, applying these methods remains difficult. Models that demonstrate excellent performance during the development phase often prove non-transferable to real-world data. Data labeling inconsistencies, mismatched available features, a limited number of accessible parameters, and incomplete sensor coverage – all of these factors undermine the "data" component, causing data-driven methods to falter. The common modeling assumption of having complete information about the network where defects are to be localized leads to a dual outcome: on one hand, accuracy on synthetic data can be outstanding, yet on the other hand, it renders training such a model for a real-world DHN practically impossible [17–21].

A common strategy to address data scarcity involves a hybrid approach: generating synthetic data through physical simulation, followed by model training on this augmented dataset [22]. This methodology enables both the evaluation of various algorithms prior to deployment and the enhancement of existing models [23]. The underlying data shortage, however, remains both at the systemic level (a lack of public datasets and benchmarks) and at the physical level, given that sensor coverage across the network pipelines is usually poor (thus, creating the need for their smart placement [24,25]), severely limiting the ability to develop robust and generalizable models.

To maximize the utility of the limited data available, a promising approach is to look at the heating network's structure. The propagation of thermal and hydraulic transients within DHNs exhibits strong spatial locality: disturbances originating at a given location primarily influence immediately adjacent network elements before attenuating or dispersing further [26]. This neighborhood-dependent behavior establishes network topology as a primary determinant of transient dynamics and, consequently, of observable fault signatures.

Accordingly, a physically consistent modeling framework must explicitly preserve these spatial interdependencies. Representing the distribution system as a graph (where

nodes correspond to junctions, consumers, or injection points, and edges represent connecting pipe segments) naturally encodes the adjacency relationships that govern wave propagation [18]. Within this representation, graph neural networks (GNNs) provide a theoretically grounded approach for fault analysis, as their message-passing mechanism inherently aggregates state information across topologically connected elements, mirroring the physical diffusion of thermal and hydraulic disturbances.

Studies on water [27–31] and gas [32] distribution systems have demonstrated that GNN-based methods achieve superior fault localization accuracy compared to topology-agnostic approaches, precisely because they exploit the structural constraints that dictate how anomalies propagate through interconnected network components. Specific results include leakage detection accuracy above 90%, topology-aware pressure reconstruction with significantly lower error compared to previously used methods such as SVM (Şahin & Yüce, 2023), Truong et al. reported a $\approx 40\%$ reduction in pressure estimation error using a GNN-based model. For gas network state estimation, Li & Wu (2024) proposed a CGNN (Composite Graph Neural Network) architecture that fuses segment-level features, reporting accuracies of 90.28% and $> 85\%$ in dense and sparse monitoring zones, respectively. Inspired by these findings, this paper investigates the applicability of GNNs to DHNs, aiming to adapt these topology-aware architectures to the specific challenges of heat distribution monitoring.

2. Materials and Methods

2.1. Overall Approach

This research introduces a two-stage methodology designed to address data scarcity and complexity in DHNs. First, to decouple topological learning from physical modeling, the design and validation of the GNN architecture are performed using a fully abstract synthetic dataset (random trees). Subsequently, this architecture is tested and adapted on data generated by a physical simulation of a real DHN, also involving data preprocessing and performance evaluation.

2.2. Stage 1: Architecture Design on Synthetic Data

The objective of this stage is to investigate and design a GNN architecture capable of effectively capturing and aggregating information from multi-step (k -hop) neighborhoods of graph edges. Since real DHN data is limited, the architecture must demonstrate this capability with minimal computational complexity and without an excessive number of parameters.

To achieve this, we generated an artificial dataset, built and conducted experiments with various neural network models, and examined how different hyperparameters and mechanisms (convolutions, attention, pooling, skip-connections, etc.) affected the performance of these models. The target variable for each edge was explicitly defined as a function of the features of nodes located within k steps from that edge (see equation 1), which simulates physical dependencies in networks. This simple and well-defined synthetic dataset was created to enable a fully controlled experiment for architecture selection.

To formulate a regression task on graph edges, we designed a dataset as a set of random trees. The node features in a tree included a random scalar value and the two-dimensional coordinates (x, y) of the node in an arbitrary planar embedding of the graph on the plane. The edge features consisted only of random scalar values. The target variable for each edge e was defined as a function $f_k(e)$:

$$f_k(e) = \sum_{i:w_i \in N_k(e)} X_i \cdot \lambda(w_i, e), \quad (1)$$

where $N_k(e)$ is the set of nodes within a k -hop neighborhood of edge e , $X_i \in [0, 1]$ is the node attribute, and λ is a decay factor.

We tested several different decay factors, including:

- $\lambda = 1$ – constant decay,
- $\lambda = 2^{-d}$, $d = \arg \min_k \{w_i \in N_k(e)\}$ – distance-based exponential decay,
- $\lambda = \prod_{w_i \in P} W_i$, where P is the set of all edges in the shortest path between edge w_i and edge e (excluding e), and $W_i \in [0.1, 1]$ is edge feature of w_i – random exponential decay of the node feature weight based on the distance.

An example with $\lambda = 1$ is illustrated in Figure 1.

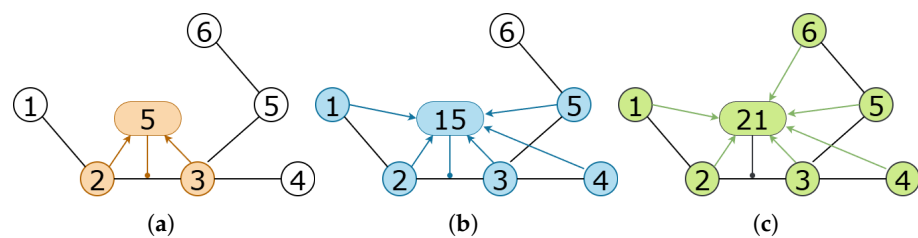


Figure 1. Illustration of target variable behavior across different k : (a) $k = 0$. (b) $k = 1$. (c) $k = 2$.

2.3. Proposed Architecture.

Experiments were conducted with various GNN architectures, including different kinds of message-passing, GCN, GAT, and GraphSAGE [33–35] layers, as well as regularization methods such as Dropout, LayerNorm, and GraphNorm. Ultimately, an architecture utilizing an attention mechanism was proposed, demonstrating strong performance. The constructed GNN consists of two main components:

1. A **node encoder** using a sequence of GATv2 layers, LayerNorm, and ReLU activation, aggregated via Jumping Knowledge with concatenation [36,37].
2. An **edge attention layer** implementing multi-head attention over edges sharing a common node, followed by LayerNorm and skip-connections.

The main findings from the architecture search and ablation study were: first, the use of spatial node coordinates (x, y) significantly improved prediction quality; second, the attention mechanism was a key component regardless of the complexity of the task; and third, the selected architecture outperformed all others considered in learning local functional dependencies.

2.4. Physically Modeled Dataset Generation

To generate the dataset required for training the GNN, a thermohydraulic simulation was employed to model the physical behavior of the DHN under various operational and fault conditions. The heating network in the simulation is represented as an oriented graph, where the nodes correspond to pipe junctions and heat sources, and the edges represent pipeline branches with specified geometric, thermal, and hydraulic parameters. Notably, to close the hydraulic circuit, the end consumers are modeled as specialized edges connecting the supply and return sub-networks; these consumer edges are assigned specific local hydraulic resistance coefficients and heat transfer parameters to simulate the pressure drop and thermal extraction of building's internal heating system. The flow distribution within the network is governed by the hydraulic circuit theory, requiring the simultaneous solution of mass and momentum conservation equations [38].

The hydraulic state is determined from conservation of mass at each node and a momentum (head-loss) relation on each branch. Let branch $b = (i, j)$ connect nodes i and j ,

and let its positive direction be chosen from i to j . The unknown hydraulic variables are the nodal pressures p_i and the branch mass flow rates m_b , whose sign indicates the flow direction relative to the branch orientation. In discrete graph form, nodal continuity is written as

$$\sum_{b \in \mathcal{U}_i} \varepsilon_{ib} m_b = q_i, \quad (2)$$

where $\mathcal{U}_i$ is the set of branches incident to node i , $\varepsilon_{ib} = \pm 1$ is the incidence sign determined by branch orientation, and q_i represents the external mass source or sink at the node.

The momentum conservation along each pipeline segment is described by the nonlinear relation

$$s_b |m_b| m_b = -\Delta p_b, \quad (3)$$

where $\Delta p_b = p_j - p_i$ is the pressure drop across branch b , and s_b is the effective hydraulic resistance coefficient of the branch. In general, s_b is determined by the hydraulic properties of the branch and may include both distributed friction and local hydraulic losses. In this case, the effective resistance can be written as

$$s_b = \frac{1}{2\rho_b A_b^2} \left(\lambda_b \frac{L_b}{D_b} + \zeta_b \right), \quad (4)$$

where λ_b is the Darcy friction factor, L_b is the branch length, D_b is the hydraulic diameter, A_b is the cross-sectional area, and ζ_b is the sum of local hydraulic resistance coefficients.

To solve these equations, the simulation employs a SIMPLE-like iterative procedure. This algorithm utilizes an explicit equation for pressure correction,

$$\nabla \cdot (\kappa \nabla p') = \nabla \cdot m - q, \quad (5)$$

where p' is the pressure correction and

$$\kappa_b = \frac{1}{s_b |m_b|}. \quad (6)$$

The solver sequentially updates the nodal pressure field and branch mass flow rates through iterations until strict mass-balance convergence is achieved across the entire network.

Thermal calculations are coupled to the hydraulic solution by evaluating convective heat transport and thermal losses to the environment after hydraulic calculation iteration. Heat loss from a pipe to ambient air is modeled by the Newton–Richmann relation

$$Q = \alpha F (t_b - t_{out}), \quad (7)$$

where α is the overall heat transfer coefficient, F is the pipe surface area, t_b is the average water temperature within the pipe, and t_{out} is the ambient outdoor temperature. The average pipe temperature t_b is calculated from the steady one-dimensional energy balance with distributed heat loss,

$$t_b = t_{out} + \frac{\dot{m}_b c_p}{\alpha F} (t_1 - t_{out}) \left[1 - \exp\left(-\frac{\alpha F}{\dot{m}_b c_p}\right) \right], \quad (8)$$

where $\dot{m}_b$ is the branch mass flow rate, c_p is the specific heat capacity, and t_1 is the inlet temperature of the pipe. At network nodes where multiple fluid streams intersect, perfect adiabatic mixing is assumed, and the resulting nodal temperature is computed as the mass-weighted average of the incoming flows. All thermal properties of the heat carrier fluid, including density, dynamic viscosity, and specific heat capacity, are dynamically updated during the iterations based on the local fluid temperature.

To unify the hydraulic and thermal calculations, an iterative steady-state simulation algorithm is used. The computational sequence starts by assigning initial approximations of nodal pressures and temperatures from the network boundary conditions. Based on these values, temperature-dependent thermophysical properties of the fluid are evaluated and initial branch mass flow rates are estimated. The core of the algorithm then enters an iterative hydraulic loop: the branch mass flows are updated from an iterative solution of the momentum conservation equations, after which a global pressure-correction equation is solved to adjust the nodal pressure field and correct the branch flows.

Once the hydraulic flow distribution is stabilized, the algorithm proceeds to the thermal step and updates the temperature field; the fluid density, viscosity, and heat capacity are then recalculated for each pipe using the newly obtained local temperatures. The coupled iterations continue until the predefined convergence criterion is satisfied; otherwise, the solver returns to the hydraulic phase with updated thermophysical properties.

Using this simulation approach, a local segment of a real urban DHN was modeled, comprising a central heat source, consumer stations, and the interconnecting supply and return pipeline infrastructure. The topology includes 187 nodes and 202 pipe segments. The heat source is represented by boundary conditions: specified supply flow rate and temperature, and fixed return pressure, while the boiler outlet pressure is obtained within the solution.

To provide a representative range of operating conditions, simulations followed an operational temperature curve that maps outdoor ambient temperature to the required source supply temperature and mass flow rate. Consumer heat demands were also scaled according to the same curve. The outdoor temperature was varied from -34°C to $+10^{\circ}\text{C}$ with 1°C increments to define baseline environmental scenarios.

For each baseline scenario, a grid of operational states with various modifications of baseline scenario, was generated, resulting in a final dataset of 42570 samples. Natural data noise (environmental fluctuations and measurement uncertainty) was simulated by applying random variations within 5% of heat transfer coefficient α on a randomly selected 10–20% subset of pipes. To emulate localized thermal degradation (e.g., insulation failure or pipe flooding), a set of defect scenarios in which α was increased in a single pipeline segment was generated. The defect location was swept over all pipeline segments in the network, excluding consumer-modeling pipes. Five defect severity levels were considered, corresponding to 10–20% of the baseline α value.

2.5. Stage 2: Application to DHN Fault Localization

Stage 2 builds on the design choices and assumptions that worked well in Stage 1. In Stage 1, the objective was to verify the model's capacity to aggregate information from multi-step (k -hop) neighborhoods by learning functional dependencies from synthetic data (Equation 1). This capability is critical for DHN fault localization because physical disturbances, such as pressure drops or temperature anomalies, propagate spatially through the network. Consequently, the model must capture these local topological dependencies to accurately trace the fault source.

Having understood what a suitable GNN should look like, we moved on to fault localization in the DHN. Although the simulation provides data for all 187 nodes, we deliberately zeroed out sensor readings in the majority of nodes, retaining pressure and temperature measurements at only 30 nodes to mirror real-world conditions where sensor coverage is sparse. Since edge-level regression is impractical under such sparsity, we reformulated the problem as multi-class classification. For this purpose, we adapted the architecture by replacing the final regression layer with a classification head and introducing global pooling.

We used a prepared dataset of 42,570 simulated operational states of a DHN, represented as a graph with 187 nodes (junctions, consumers, source/sink) and 202 edges (pipes) illustrated in Figure 2.

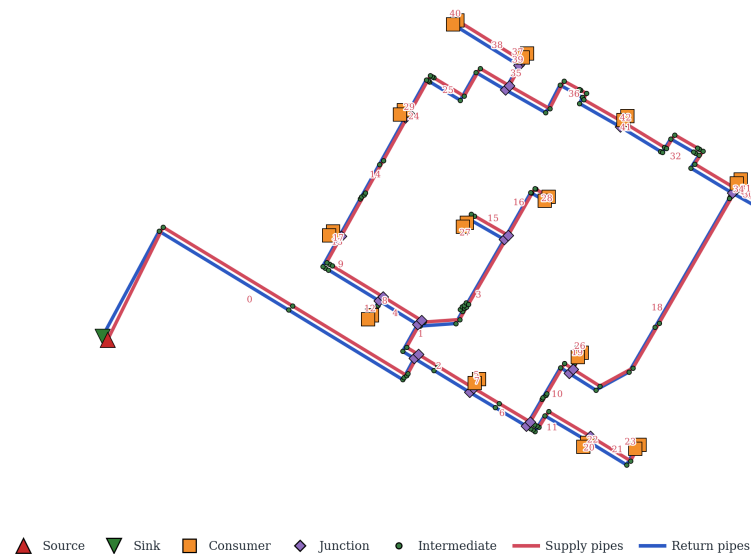


Figure 2. An illustration of the district heating network.

The following node features were used for model training:

- Local x, y coordinates of the node.
- Node type – source, sink, or pipe junction.
- Pressure at the node – a dynamic parameter.
- Temperature at the node – a dynamic parameter.

The following pipe (edge) features were used:

- Pipe diameter.
- Pipe length.
- Pipe type – supply, return, or consumer line.

Additionally, the outdoor air temperature was used as an input parameter.

The data preprocessing pipeline consists of the following steps:

1. Splitting the large heating network graph into two intersecting subgraphs – the supply and return subnets. These are formed by selecting all supply pipes, return pipes, and consumer pipes along with their incident vertices in the graph. These components have similar structures, as supply and return pipes are physically co-located.
2. Loading the data required for training, including global parameters such as outdoor air temperature and boiler plant temperatures, which are duplicated into separate tables.
 - 2.1 A numerical identifier is added to each edge in the graph, indicating which graph section the edge belongs to after partitioning. For each graph, a class (numerical label) is assigned – the index of the section containing the defective edge (or a separate class if no defects are present).
 - 2.2 The partitioning logic is as follows: edges belong to the same section if and only if they form a direct path between two key vertices. A vertex is considered a key vertex if it is a consumer vertex (incident to consumer pipes), a boiler plant vertex (heat source or sink), or a junction vertex (with in-degree + out-degree > 2).

- 2.3 Adding dynamic parameters (readings from temperature and pressure sensors) to the graph vertices corresponding to consumers and the boiler plant (30 out of 187 vertices in total).
3. Transforming the raw data into a set of tabular representations.
4. Estimating normalization parameters and scaling three groups of features (the two heating network components and the global parameters) to zero mean and unit standard deviation based on the training set.
5. Applying normalization to all tabular data.
6. Converting the normalized tabular data into graph objects used by the model.

The core architecture from Stage 1 was adapted for classification as illustrated in Figure 3:

- The final edge feature representations, enhanced by the node encoder and edge attention blocks, were concatenated with their transformed versions from all attention layers.
- A global pooling layer aggregated edge features into a single graph-level representation.
- A linear classifier layer mapped this representation to logits over the classes.

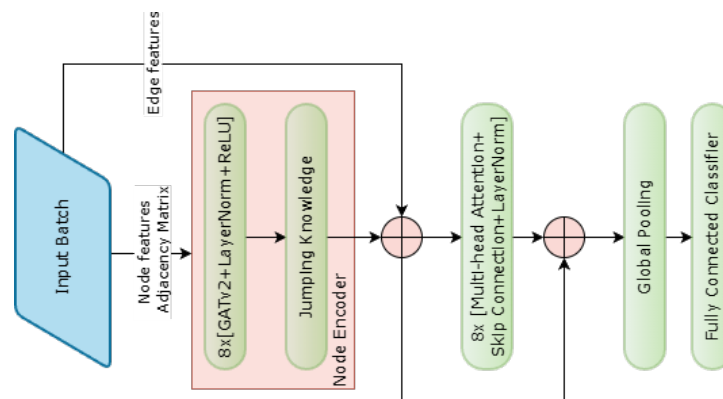


Figure 3. An illustration of the architecture of our GNN.

2.6. Training.

The model was trained for 100 epochs with a batch size of 16 graph pairs. The dataset was split into training, validation, and test sets with a ratio of 70:15:15.

The model was optimized using Adam. The initial learning rate was set to 10^{-3} , with $\beta = (0.9, 0.99)$, $\epsilon = 10^{-8}$, and weight decay $\lambda = 10^{-6}$. The learning rate was reduced exponentially each epoch down to 10^{-6} using a StepLR scheduler with a step size of 1 and $\gamma = (10^{-6}/10^{-3})^{1/100} \approx 0.933254$.

The loss function was multi-class cross-entropy (CrossEntropyLoss) with label smoothing ($\epsilon = 0.1$) and class weighting applied. The model hyperparameters are listed in Table 1.

Parameter	Value
Node encoder layer dimension	128
Edge attention layer dimension	128
Number of node encoder layers	8
Number of edge attention layers	8
Number of attention heads	8
Jumping knowledge mode	concatenation

Table 1. Model Hyperparameters.

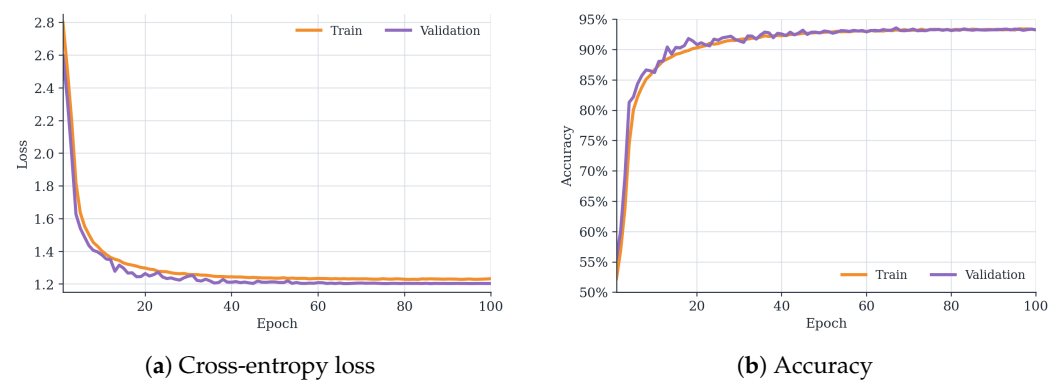


Figure 4. Training vs. validation performance over epochs.

2.7. Implementation

All models were implemented using Python 3.12, PyTorch 2.4.1, and the PyTorch Geometric 2.6.1 library.

3. Results and Discussion

3.1. Main Results

The performance of the proposed GNN architecture was evaluated using the physically modeled dataset described in Section 2.5. To ensure a robust assessment of generalization, the dataset was randomly partitioned into training, validation, and test sets in a 70:15:15 ratio. The model was trained for 100 epochs using the Adam optimizer, with convergence observed after approximately 50 epochs as indicated by the stabilization of the validation loss on Figure 4.

Table 2 presents the results achieved by our model on the test dataset. The reported number of classes (30) includes both the specific fault locations across the network sections and an additional class representing the normal operating state ("no defect"). The mean Accuracy and Macro F1 scores exceeding 90% demonstrate that the model exhibits strong predictive performance on sparse data, successfully capturing graph dependencies and effectively handling the classification task.

Metric	Value on supply subnet	Value on return subnet
Accuracy	0.960	0.910
Macro F1	0.930	0.869
Weighted F1	0.957	0.908
Number of classes	30	30
Number of samples	6380	6380

Table 2. Performance metrics of the proposed model on the test set.

The observed performance discrepancy between the supply (0.960) and return (0.910) subnets aligns with the physical characteristics of the system. The return pipeline has a narrower range of dynamic values (temperature/pressure). Since the fluid has already cooled and pressure has dropped, the relative impact of a defect is smaller compared to the background noise.

To interpret the model’s errors, we aimed to understand how far the predictions deviate from the ground truth when the model misclassifies (in terms of graph distance). For this purpose, we introduce a "distance confusion matrix" in Figure 5. The x-axis represents the true fault location (graph section), while the y-axis shows the topological distance between the predicted and true sections. Each cell indicates the proportion of

predictions falling at a given distance from the true class. Perfect predictions would yield a vertical line at distance 0 with value 1.0.

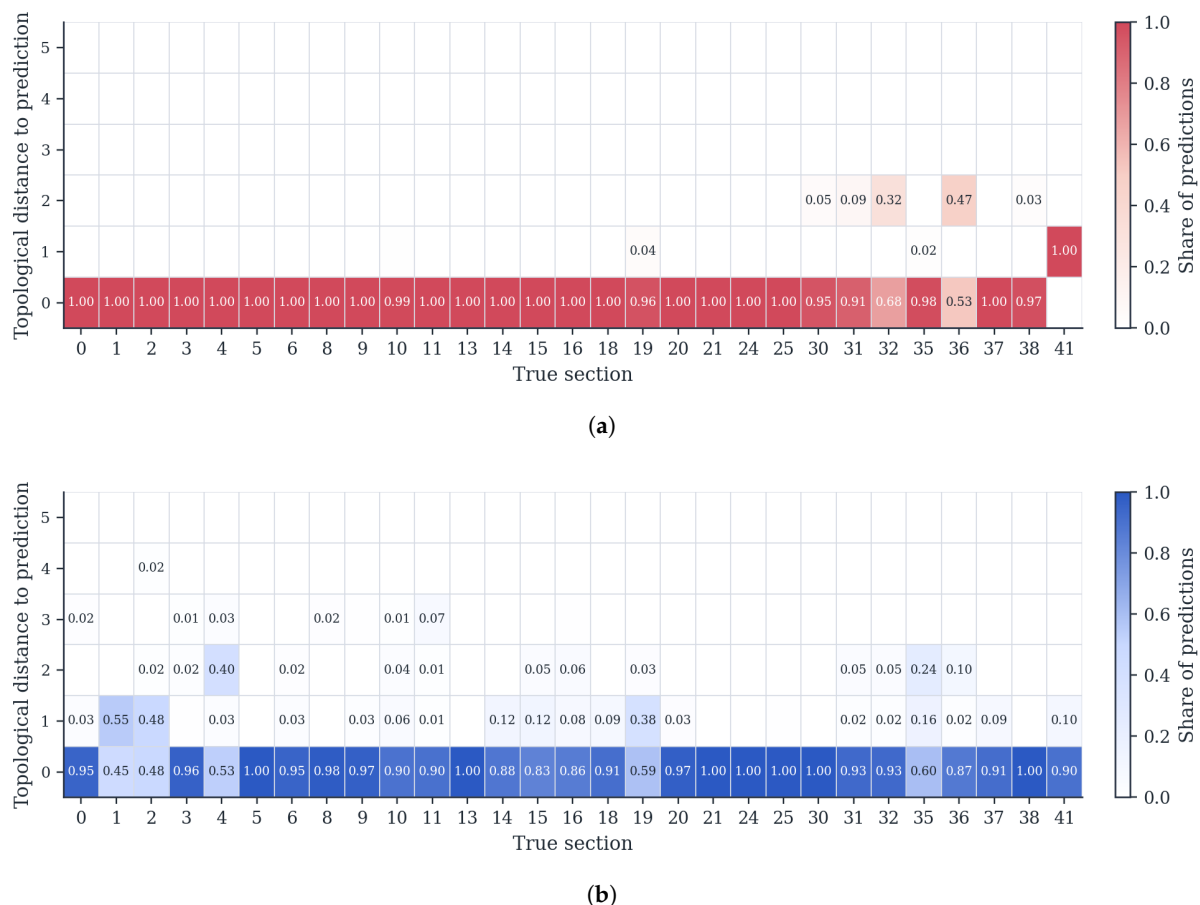


Figure 5. Distance confusion matrices: (a) Supply subnet. (b) Return subnet.

Notably, the distribution of errors in Figure 5 demonstrates a strong bias towards local mistakes. We observe that the error mass is heavily concentrated within a topological distance of 1–2 sections, with negligible occurrences of errors beyond that distance (relative to the network diameter of 12). Because heat loss and pressure drop affect neighboring segments, the model correctly identifies the neighborhood of the fault even if it misassigns the specific pipe edge. From an operational standpoint, this suggests that even in cases of misclassification, the model effectively narrows the search area to the vicinity of the actual fault.

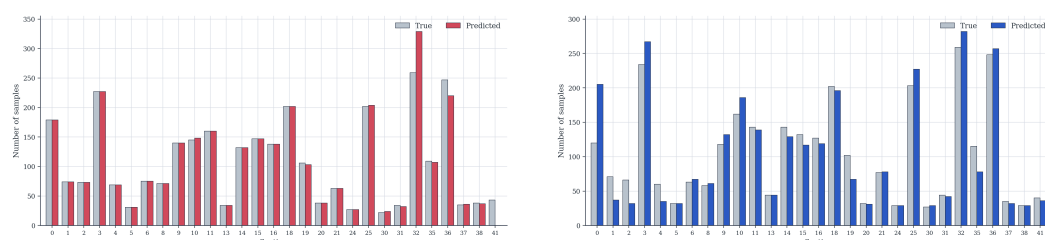


Figure 6. Distribution of predicted classes for (a) supply and (b) return subnets, excluding "no failure" class.

The distribution of predicted classes for each subnet is shown in Figure 6.

3.2. Ablation Studies

To validate our architectural choices, we conducted a comprehensive ablation study examining three key aspects: (1) model depth and the impact of multi-head attention mechanisms with skip connections, (2) pooling strategies, (3) number of attention heads, and (4) choice of graph convolution layer type. All experiments were performed on the fault classification task using the DHN dataset described in Section 2.5.

As can be seen in Tables 3, 4, the ablation experiments confirm our hypothesis that the proposed architecture is optimal among comparable alternatives, with the attention mechanism serving as a key factor in prediction quality, both in the node encoder and as an edge-attention layer.

Configuration	S (4 layers)		M (8 layers)		L (16 layers)	
	Acc	F1	Acc	F1	Acc	F1
Attention + skip connections	91.4	86.8	92.1	87.6	73.5	30.0
Attention w/o skip connections	88.7	79.2	90.2	75.7	73.5	31.5
No attention (MLP)	84.9	67.3	86.6	76.5	71.3	24.8
Number of parameters	725K		1.45M		2.9M	

Table 3. Ablation study results: impact of model depth, multi-head attention mechanism, and skip connections on classification accuracy (%) and macro F1-score (%).

Configuration	Accuracy (%)	Macro F1 (%)
Baseline: 8 layers, GATv2, attention + skip, 4 heads, attention pooling		
Pooling strategies		
Attention pooling	92.1	87.6
Mean pooling	86.7	65.9
Attention heads		
2 heads	88.7	78.5
4 heads	92.1	87.6
8 heads	93.6	90.0
16 heads	91.8	86.1
Graph convolution layer types		
GATv2	92.1	87.6
GraphSAGE	83.7	52.6
GCN	66.2	16.6

Table 4. Ablation study results on the 8-layer architecture with attention and skip connections.

Future Perspectives. While the unified architecture demonstrated robust performance, the observed discrepancy in accuracy between the supply (0.960) and return (0.910) subnets suggests that the current modeling approach may not fully capture the distinct hydraulic regimes. To address this, future work will explore a two-stage extension of the current framework.

First, we will assess the potential benefits of decoupling the networks by training two distinct specialized models. This baseline experiment could help establish the performance ceiling achievable by treating the supply and return domains as separate learning problems. Furthermore, it may be beneficial to implement graph-type conditioning, where the message-passing mechanism explicitly utilizes different weight matrices for supply and return features. Such an approach might allow the model to better distinguish the patterns in data, potentially closing the accuracy gap while maintaining a single deployable system.

4. Conclusions

In this study, we addressed the challenge of fault localization in DHNs under conditions of data scarcity, a common constraint in real-world infrastructure monitoring. By using GNNs, we exploited the topological structure of DHNs to overcome the limitations of traditional physics-based and topology-agnostic data-driven methods.

The research was conducted in two distinct stages. First, we utilized a synthetic dataset to identify an architecture capable of capturing long-range dependencies in graph-structured data. This led to the design of a deep GNN architecture integrating GATv2 node encoders and edge-attention mechanisms. Second, we validated this architecture on a physically modeled dataset generated via thermohydraulic simulations, representing a realistic urban heating network.

The results demonstrated that the proposed model achieves high localization accuracy on the test set: 96.0% for the supply subnet and 91.0% for the return subnet, with corresponding Macro F1 scores of 93.0% and 86.9%, respectively. These metrics were obtained under sparse sensor coverage (only 30 of 187 nodes contained non-zero thermal and pressure features), simulating realistic, data-limited conditions. The ablation studies further confirmed that the attention mechanism and network depth are critical factors for performance. The distance confusion matrix reveals that most errors are localized within a topological distance of 1–2 sections from the actual defect. This confirms that the model implicitly captures the attenuation of hydraulic and thermal disturbances along the network topology.

These findings suggest that topology-aware deep learning offers a promising approach for DHN monitoring. Future work will focus on validating the approach on historical field data from operational networks and testing it across different district heating geometries, while addressing the inevitable domain shift between simulated training regimes and real-world seasonal and load variations. However, we recognize that the most significant challenge remains the acquisition of high-quality, labeled data to support such data-driven methodologies.

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

CGNN	Composite Graph Neural Network
CNN	Convolutional Neural Network
DHN	District Heating Network
GAT	Graph Attention Network
GCN	Graph Convolutional Network
GNN	Graph Neural Network
GraphSAGE	Graph SAmple and aggreGatE
ReLU	Rectified Linear Unit
SCADA	Supervisory Control and Data Acquisition
SIMPLE	Semi-Implicit Method for Pressure-Linked Equations
SVM	Support Vector Machine
UAV	Unmanned Aerial Vehicle

Nomenclature

A	Cross-sectional area of the pipe
c_p	Specific heat capacity of the fluid
d	Distance metric
e	Graph edge
F	Surface area of the pipe
$f_k(e)$	Target variable function for edge e
k	Number of hops (neighborhood radius)
m_l	Mass flow rate vector along the connected edges
$N_k(e)$	Set of nodes within a k -hop neighborhood of edge e
p	Pressure
p'	Pressure correction
P	Set of edges in the shortest path between nodes
q	Mass source or sink at the specific node
Q	Heat loss from pipe to ambient air
s	Hydraulic resistance coefficient
t_1	Inlet temperature
t_b	Average water temperature within the pipe
t_{out}	Ambient outdoor temperature
t_i	Nodal temperature
u_b	Flow velocity
W_i	Edge feature
X_i	Node attribute
α	Overall heat transfer coefficient
Δp	Pressure drop over the pipe
κ	Coefficient derived from hydraulic resistance ($1/(s m_l)$)
λ	Decay factor
ρ	Fluid density
$\nabla \cdot$	Divergence operator

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